

ASHISH RANJAN

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SUMMARY

Backend-focused Software Engineer with hands-on experience in building secure and scalable REST APIs using Java, Spring Boot, and Spring Security. Skilled in backend architecture, authentication/authorization, database-driven applications, and production-oriented API development. Additionally experienced in Generative AI systems, including RAG pipelines, vector databases, and AI-powered backend integrations.

TECHNICAL SKILLS

Programming Languages: Java, Python, SQL, C/C++

Backend Development: Spring, Spring Boot, REST APIs, Spring Security, Spring Data JPA, JUnit, Mockito

Databases: PostgreSQL, MySQL, MongoDB, Vector Databases

AI/ML: Generative AI, Agentic AI, RAG Pipelines, Scikit-learn, LangChain

Tools: Git, GitHub, Postman, Docker

CS Fundamentals: Data Structures and Algorithms, Object Oriented Programming, Operating Systems, Database Management Systems, Computer Networks

EXPERIENCE

HCLTech

Sept 2024 – Present

Software Engineer

Bengaluru, India

- Developed Generative AI prototypes (POCs) using Python, implementing RAG pipelines with embeddings and vector databases (FAISS, Pinecone).
- Built and exposed RESTful APIs to serve AI/ML functionalities, enabling structured querying over unstructured datasets.
- Utilized frameworks like LangChain and LlamaIndex to create modular, reusable pipelines for rapid experimentation.
- Designed and debugged REST APIs, implementing request validation, error handling, and service-layer logic in an Agile environment.
- Built backend systems using Java and Spring Boot, implementing authentication, role-based access control, and production-grade backend architecture.

PROJECTS

Video Rental Management System | [GitHub](#)

- Developed a secure RESTful backend system for an online video rental platform, enabling customers to browse, rent, and return videos while supporting admin-controlled video management workflows.
- Implemented JWT-based authentication and role-based authorization using Spring Security, with protected APIs, BCrypt password encryption, and access control for CUSTOMER and ADMIN roles.
- Designed scalable REST APIs with layered architecture, DTO pattern, centralized exception handling, pagination, sorting, filtering, and business validations including rental limits and video availability management.
- **Tech Stack:** Java, Spring Boot, Spring Security, JWT, JPA/Hibernate, MySQL, Gradle, REST APIs, Swagger/OpenAPI, Git, Postman

Task Management API | [GitHub](#)

- Developed a secure and scalable Task Management REST API using Spring Boot with layered architecture, enabling task creation, updating, deletion, and user-specific task management functionalities.
- Implemented authentication and authorization using Spring Security and JWT-based security, including password encryption, stateless session handling, and protected REST endpoints.
- Designed and integrated database operations using JPA/Hibernate with optimized entity relationships and repository abstraction, following clean backend engineering practices and RESTful API standards.
- **Tech Stack:** Java, Spring Boot, Spring Security, JWT, JPA/Hibernate, Gradle, MySQL, REST APIs, Git, Postman

QEats | GitHub

- QEats is a food ordering application that allows users to browse restaurants and place food orders from nearby locations.
- Built multiple backend modules using Spring Boot and developed REST APIs for restaurant discovery and order management functionalities.
- Improved application performance under high-load scenarios and implemented scalable backend components for efficient data processing.
- **Tech Stack:** Java, Spring Boot, MongoDB, REST APIs, Redis, Scientific Debugging, Git

QMoney | GitHub

- Developed a visual stock portfolio analyzer that helps users evaluate portfolios and generate investment insights using market data.
- Implemented the core portfolio management logic and designed reusable backend components following clean coding practices.
- Refactored the application to support multiple stock quote providers and improved extensibility of external API integrations.
- **Tech Stack:** Java, Spring Boot, REST APIs, Jackson, Exception Handling, Git

EDUCATION

IIIT Allahabad	06 2020 – 06 2024
<i>B.Tech - CGPA - 8.37</i>	<i>Prayagraj, India</i>
Param Gyan Niketan	06 2018 – 06 2020
<i>Class XII - Percentage - 95.2</i>	<i>Gaya, India</i>